

Putnam 2018

A1 Find all ordered pairs (a, b) of positive integers for which

$$\frac{1}{a} + \frac{1}{b} = \frac{3}{2018}.$$

Answer. There are 6 such pairs, obtained by $(1009, 2018)$, $(673, 2018 \times 673)$, $(674, 337 \times 1009)$ and their inverses.

Solution. Since both a and b are positive, we can multiply both sides by $2018ab$ to get $2018a + 2018b = 3ab$. Even better, this is the same as $(3a - 2018)(3b - 2018) = 2018^2$. Since both $3a - 2018$ and $3b - 2018$ are integers congruent to 1 (mod 3), it suffices to find the ways to decompose $2018^2 = cd$ with each c, d are 1 (mod 3). We also know that $2018^2 = 2^2 \times 1009^2$ and both 2 and 1009 are primes, so each positive divisor of 2018^2 must be in the form of $2^k \times 1009^\ell$ with $0 \leq k, \ell \leq 2$. The modulo 3 constraint also forces k to be even, so $(c, d) = (1, 2018^2), (4, 1009^2), (1009, 4 \times 1009)$ and their reverses, giving the answer as above (by letting $a = \frac{c+2018}{3}$ and $b = \frac{d+2018}{3}$).

A2 Let $S_1, S_2, \dots, S_{2^n-1}$ be the nonempty subsets of $\{1, 2, \dots, n\}$ in some order, and let M be the $(2^n - 1) \times (2^n - 1)$ matrix whose (i, j) entry is

$$m_{ij} = \begin{cases} 0 & \text{if } S_i \cap S_j = \emptyset; \\ 1 & \text{otherwise} \end{cases}$$

Calculate the determinant of M .

Answer. 1 if $n = 1$, and -1 if $n > 1$. Alternatively, $(-1)^{(n>1)}$ in the computer science terms (where false is 0 and true is 1).

Solution. If $n = 1$ then the matrix is (1) , so the determinant is 1. Let M_n be the matrix when the size of the matrix is $n \times n$, and we will show that $\det(M_{n+1}) = -(\det(M_n))^2$.

First, we use the well-known fact that swapping any two rows (and similarly swapping any two columns) negates the determinant of a square matrix. Hence, if we swap the roles of any S_i and S_j , this is as if we swap rows i and j , and then columns i and j , of the matrix M , therefore $\det(M)$ remains the same. Thus $\det(M)$ is independent of the permutations of $S_1, \dots, S_{2^n-1}$, and it suffices to solve this problem for one particular arrangement of $S_1, \dots, S_{2^n-1}$. We do the following: given the index $n + 1$, let $S_1, \dots, S_{2^n-1} \subset \{1, 2, \dots, n\}$, $S_{2^n} = \{n + 1\}$ and $S_{i+2^n} = S_i \cup \{n + 1\}$ for $i = 1, 2, \dots, 2^n - 1$. In particular, our matrix (name it M_{n+1}) looks like the following:

$$\begin{pmatrix} M_n & 0_{(2^n-1) \times 1} & M_n \\ 0_{1 \times (2^n-1)} & 1_{1 \times 1} & 1_{1 \times (2^n-1)} \\ M_n & 1_{(2^n-1) \times 1} & 1_{(2^n-1) \times (2^n-1)} \end{pmatrix}$$

Subtracting rows $2^n + 1, \dots, 2^{n+1} - 1$ by row 2^n , and then columns $2^n + 1, \dots, 2^{n+1} - 1$ by column 2^n , we end up with

$$\begin{pmatrix} M_n & 0_{(2^n-1) \times 1} & M_n \\ 0_{1 \times (2^n-1)} & 1_{1 \times 1} & 0_{1 \times (2^n-1)} \\ M_n & 0_{(2^n-1) \times 1} & 0_{(2^n-1) \times (2^n-1)} \end{pmatrix}$$

Now this is the reverse block matrix (with the two M_n 's reversed), so the determinant is now $(-1)^{2^n-1}(\det(M_n))^2 = -(\det(M_n))^2$, as desired (with the $2^n - 1$ due to the size of M_n).

A3 Determine the greatest possible value of $\sum_{i=1}^{10} \cos(3x_i)$ for real numbers $x_1, x_2, \dots, x_{10}$ satisfying $\sum_{i=1}^{10} \cos(x_i) = 0$.

Note: Solution featured on America Mathematical Monthly!

Answer. $\frac{480}{49}$.

Solution. For any real number x ,

$$\begin{aligned}\cos(3x) &= \cos(2x)\cos x - \sin(2x)\sin x = (2\cos^2 x - 1)\cos x - (2\sin x \cos x)\sin x \\ &= 2\cos^3 x - \cos x - 2(1 - \cos^2 x)\cos x = 4\cos^3 x - 3\cos x.\end{aligned}$$

Thus, whenever $\sum_{i=1}^{10} \cos x_i = 0$, we have

$$\sum_{i=1}^{10} \cos 3x_i = \sum_{i=1}^{10} (4\cos^3 x_i - 3\cos x_i) = 4\sum_{i=1}^{10} \cos^3 x_i$$

Set $y_i = \cos x_i$. Suppose that k of the ten numbers y_i 's are positive. Without loss of generality, we may assume that y_i is positive if and only if $1 \leq i \leq k$. For a given y , we can find a real number x with $y = \cos x$ if and only if $|y| \leq 1$. So we wish to maximize

$$4\sum_{i=1}^{10} y_i^3 = 4\left(\sum_{i=1}^k y_i^3 - \sum_{i=k+1}^{10} |y_i|^3\right)$$

given

$$|y_i| \leq 1 \text{ for all } i, \sum_{i=1}^{10} y_i = 0, y_1, \dots, y_k \geq 0, \text{ and } y_{k+1}, \dots, y_{10} \leq 0$$

Let $\sum_{i=1}^k y_i = S$. Because $0 \leq y_i \leq 1$ for $1 \leq i \leq k$, we have $\sum_{i=1}^k y_i^3 = S$. On the other hand, we have

$$0 = \sum_{i=1}^{10} y_i = S - \sum_{i=k+1}^{10} |y_i|, \text{ so } \sum_{i=k+1}^{10} |y_i| = S$$

The power mean inequality then yields

$$\sqrt[3]{\frac{\sum_{i=k+1}^{10} |y_i|^3}{10-k}} \geq \frac{\sum_{i=k+1}^{10} |y_i|}{10-k} = \frac{S}{10-k}, \text{ so } \sum_{i=k+1}^{10} |y_i|^3 \geq \frac{S^3}{(10-k)^2}$$

Therefore,

$$4\sum_{i=1}^{10} y_i^3 \leq 4\left(S - \frac{S^3}{(10-k)^2}\right)$$

The right side of the inequality has derivative $4(1 - \frac{3S^2}{(10-k)^2})$ as a function of S and so is increasing when $0 \leq S \leq \frac{10-k}{\sqrt{3}}$ and decreasing beyond that. On the other hand, because $y_i \leq 1$ for all i with $1 \leq i \leq k$, we have $S \leq k$. When $k \leq 3$, we have $k \leq \frac{10-k}{\sqrt{3}}$, and therefore

$$4\left(S - \frac{S^3}{(10-k)^2}\right) \leq 4\left(k - \frac{k^3}{(10-k)^2}\right)$$

while when $k \geq 4$ we get the estimate

$$4\left(S - \frac{S^3}{(10-k)^2}\right) \leq 4\left(\frac{10-k}{\sqrt{3}} - \frac{((10-k)/\sqrt{3})^3}{(10-k)^2}\right) = \frac{8}{3\sqrt{3}}(10-k) \leq \frac{8}{3\sqrt{3}} \cdot 6 = \frac{16}{\sqrt{3}}$$

Comparing the values of $4\left(k - \frac{k^3}{(10-k)^2}\right)$ for $k = 1, 2, 3$, we find that the largest value is $\frac{480}{49}$ when $k = 3$. This is greater than $\frac{16}{\sqrt{3}}$. So we have shown that

$$\sum_{i=1}^{10} \cos 3x_i \leq \frac{480}{49} \text{ whenever } \sum_{i=1}^{10} \cos x_i = 0$$

To finish, note that the value $\frac{480}{49}$ can be attained by taking

$$x_1 = x_2 = x_3 = 0 \text{ and } x_4 = x_5 = x - 6 = x_7 = x_8 = x_9 = x_{10} = \arccos\left(-\frac{3}{7}\right)$$

A4 Let m and n be positive integers with $\gcd(m, n) = 1$, and let

$$a_k = \left\lfloor \frac{mk}{n} \right\rfloor - \left\lfloor \frac{m(k-1)}{n} \right\rfloor$$

for $k = 1, 2, \dots, n$. Suppose that g and h are elements in a group G and that

$$gh^{a_1}gh^{a_2}\dots gh^{a_n} = e,$$

where e is the identity element. Show that $gh = hg$. (As usual, $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x .)

Solution. (Modified from my post on AoPS). We first notice the following relation:

$$\begin{aligned} \frac{m}{n} - 1 &= \frac{mk}{n} - \frac{m(k-1)}{n} - 1 \\ &\leq \left\lfloor \frac{mk}{n} \right\rfloor - \frac{m(k-1)}{n} \\ &\leq a_k = \left\lfloor \frac{mk}{n} \right\rfloor - \left\lfloor \frac{m(k-1)}{n} \right\rfloor \\ &\leq \frac{mk}{n} - \left\lfloor \frac{m(k-1)}{n} \right\rfloor \\ &< \frac{mk}{n} - \frac{m(k-1)}{n} + 1 \\ &= \frac{m}{n} + 1 \end{aligned}$$

Thus $\lfloor \frac{m}{n} \rfloor \leq a_k \leq \lceil \frac{m}{n} \rceil$ for each a_k .

Now we proceed by inducting on both m and n . The case where $m = 1$ gives rise to the situation $a_k = 0$ for all $k < n$ and $a_n = 1$ so we have $g^n h = e$. Since $h = g^{-n}$, $gh = hg = g^{-(n-1)}$, and thus commute. Similarly, the case where $m = 2$ gives rise to the $a_1 = m$ so $gh^n = e$, and thus $g = h^{-n}$ and by a similar logic, g commutes with h .

Now fix the pair (m, n) and suppose that the desired statement holds true for all (m', n') with $m' \leq m$ and $n' \leq n$ and $(m', n') \neq (m, n)$, and $\gcd(m, n) = 1$. We split this into two cases, bearing in mind that $(m, n) \neq (1, 1)$ here (otherwise our base cases take care of it).

The easier is $m > n$, in which case we have $a_k \geq 1$ for all k . Let $d = \lfloor \frac{m}{n} \rfloor$ and let m' be $m' \equiv m \pmod{n}$ with $0 \leq m' < n$, so $m = dn + m'$. Now let $b_k = a_k - d$, and we can rewrite a_k in the following form:

$$\begin{aligned} a_k &= \left\lfloor \frac{mk}{n} \right\rfloor - \left\lfloor \frac{m(k-1)}{n} \right\rfloor \\ &= \left\lfloor \frac{(m' + dn)k}{n} \right\rfloor - \left\lfloor \frac{(m' + dn)(k-1)}{n} \right\rfloor \\ &= \left\lfloor \frac{m'k}{n} + dk \right\rfloor - \left\lfloor \frac{m'(k-1)}{n} + d(k-1) \right\rfloor \\ &= \left\lfloor \frac{m'k}{n} \right\rfloor - \left\lfloor \frac{m'(k-1)}{n} \right\rfloor + d \\ &= b_k + d \end{aligned}$$

so we have

$$e = gh^{a_1}gh^{a_2} \dots gh^{a_n} = gh^{b_1+d}gh^{b_2+d} \dots gh^{b_n+d} = (gh^d)h^{b_1} \dots (gh^d)h^{b_n}$$

and by our inductive hypothesis, since the pair (gh^d, h) corresponds to (m', n) and $\gcd(m', n) = \gcd(m, n) = 1$, and $m' < n < m$, gh^d and h commutes. Thus, $gh^{d+1} = gh^d h = hgh^d$, and by cancelling h^d from the right we have $gh = hg$, as desired.

The harder part is $m < n$, i.e. $\frac{m}{n} < 1$. By the lemma above, we have $0 \leq a_k \leq 1$ for each a_k . We also have $a_k = 1$ if and only if there exists an integer a such that $\frac{m(k-1)}{n} < a \leq \frac{mk}{n}$. We can rearrange this to form $a \leq \frac{mk}{n} < a + \frac{m}{n}$, which by rearranging again becomes $\frac{na}{m} \leq k < \frac{na}{m} + 1$. This gives the condition $k = \lceil \frac{na}{m} \rceil$. By omitting those a_k 's that are 0 we can rewrite the original equation as

$$e = g^{b_1}h \dots g^{b_m}h$$

with each $b_k = \lceil \frac{nk}{m} \rceil - \lceil \frac{n(k-1)}{m} \rceil$. Again let $d = \lfloor \frac{n}{m} \rfloor$ we can write $n = dm + n'$ with $n' \equiv n \pmod{m}$ and $n' < m$, and we have

$$\begin{aligned} b_k &= \left\lceil \frac{nk}{m} \right\rceil - \left\lceil \frac{n(k-1)}{m} \right\rceil \\ &= \left\lceil \frac{(dm + n')k}{m} \right\rceil - \left\lceil \frac{(dm + n')(k-1)}{m} \right\rceil \\ &= \left\lceil \frac{n'k}{m} + dk \right\rceil - \left\lceil \frac{n'(k-1)}{m} + d(k-1) \right\rceil \\ &= \left\lceil \frac{n'k}{m} \right\rceil - \left\lceil \frac{n'(k-1)}{m} \right\rceil + d \end{aligned}$$

Thus letting $c_k = b_k - d = \left\lceil \frac{n'k}{m} \right\rceil - \left\lceil \frac{n'(k-1)}{m} \right\rceil$ we have

$$e = g^{b_1}h \dots g^{b_m}h = g^{c_1}(g^d h) \dots g^{c_m}(g^d h)$$

and since $n' < m < n$ and the pair $(g, g^d h)$ corresponds to the pair (m, n') with $\gcd(m, n') = 1$, by inductive hypothesis, g and $g^d h$ commutes. Thus $gg^d h = g^d hg$, and cancelling g^d from the left we have $gh = hg$.

A5 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an infinitely differentiable function satisfying $f(0) = 0$, $f(1) = 1$, and $f(x) \geq 0$ for all $x \in \mathbb{R}$. Show that there exist a positive integer n and a real number x such that $f^{(n)}(x) < 0$.

Solution. Suppose for sake of contradiction that $f^{(n)}(x) \geq 0$ for all $n \geq 0$ and $x \in \mathbb{R}$; this would mean f (and all its derivatives) are non-decreasing, hence $f(x) = 0$ for all $x < 0$. Consequently, $f^{(n)}(x) = 0$ for all positive n and $x < 0$; and this gives $f^{(n)}(0) = 0$ for all positive integers n .

We now show that for all $x > 0$ and all integers $n > 0$, $\frac{f(x)}{x^n}$ is non-decreasing in x . Indeed, we shall show via induction on m that $\frac{f^{(n-m)}(x)}{x^m}$. Base case is where $f^{(n)}(x)$ is non-decreasing in $x > 0$. Now suppose that for some $m \geq 1$ we have $\frac{f^{(n-m+1)}(x)}{x^{m-1}}$ non-decreasing. Take $0 < x < y$, we have

$$f^{(n-m)}(x) = \int_0^x f^{(n-m+1)}(t)dt \leq f^{(n-m+1)}(x) \int_0^x \left(\frac{t}{x}\right)^{m-1} dt = \frac{f^{(n-m+1)}(x) \cdot x^m}{x^{m-1}m}$$

$$f^{(n-m)}(y) - f^{(n-m)}(x) = \int_x^y f^{(n-m+1)}(t)dt \geq f^{(n-m+1)}(x) \int_x^y \left(\frac{t}{x}\right)^{m-1} dt$$

$$= \frac{f^{(n-m+1)}(x) \cdot (y^m - x^m)}{x^{m-1}m}$$

and therefore

$$\frac{f^{(n-m)}(y)}{f^{(n-m)}(x)} \geq 1 + \frac{y^m - x^m}{x^m} = \frac{y^m}{x^m}$$

as desired.

Now going back to the problem, we now have that for all $x < 1$ and positive integer $n > 0$, we have $f(x) \leq x^n$. Consequently, taking $n \rightarrow \infty$ for any fixed $x \in (0, 1)$ we have $f(x) = 0$ for all $x < 1$. This means f is discontinuous at $x = 1$, contradicting that it is infinitely differentiable.

- A6** Suppose that A, B, C , and D are distinct points, no three of which lie on a line, in the Euclidean plane. Show that if the squares of the lengths of the line segments AB, AC, AD, BC, BD , and CD are rational numbers, then the quotient

$$\frac{\text{area}(\triangle ABC)}{\text{area}(\triangle ABD)}$$

is a rational number.

Solution. Let the perpendicular from C and D to AB be H_1 and H_2 , respectively. Then the ratio of the areas of the two triangles is $\frac{CH_1}{DH_2}$. The central claim is the following:

Lemma 1. *The lengths $AH_1 \cdot AB, AH_2 \cdot AB, BH_1 \cdot AB, BH_2 \cdot AB$ are all rational, as well as CH_1^2, DH_2^2 .*

Proof. We have $AH_1^2 + CH_1^2 = AC^2$ so $AC^2 - BC^2 = AH_1^2 - BH_1^2 \in \mathbb{Q}$. In dot product language, we have:

$$(A\vec{H}_1 - H_1\vec{B}) \cdot \vec{AB} = (A\vec{H}_1 - H_1\vec{B}) \cdot (A\vec{H}_1 + H_1\vec{B}) = AH_1^2 - H_1B^2$$

is rational. Given that $\|AB\|^2 \in \mathbb{Q}$, $\|A\vec{H}_1 - H_1\vec{B}\|^2 \in \mathbb{Q}$. Iteratively, this gives both $\|AH_1\|^2 + \|H_1B\|^2 \in \mathbb{Q}$, and $A\vec{H}_1 \cdot H_1\vec{B} \in \mathbb{Q}$. We may further deduce $\|AH_1\|^2, \|BH_1\|^2 \in \mathbb{Q}$ and so $\|AC\|^2 \in \mathbb{Q}$, we have $\|CH_1\|^2 \in \mathbb{Q}$. Finally, $A\vec{H}_1 \cdot \vec{AB} = \|AH_1\|^2 + A\vec{H}_1 \cdot H_1\vec{B} \in \mathbb{Q}$. The other set of identities (involving H_2 and D) follows similarly. $\square$

Now, since $AH_1 \cdot AB$ and $AH_2 \cdot AB$ are rational, so is the difference $H_1H_2 \cdot AB$ (in particular, $\|H_1H_2\|^2 \in \mathbb{Q}$). We now claim that both $(\|CH_1\| + \|DH_2\|)^2$ and $(\|CH_1\| - \|DH_2\|)^2$ are rational. Indeed, if H_1 and H_2 are on different sides of AB , then $(\|CH_1\| + \|DH_2\|)^2 + \|H_1H_2\|^2 = \|CD\|^2$; if they are on the same side of AB then $(\|CH_1\| - \|DH_2\|)^2 + \|H_1H_2\|^2 = \|CD\|^2$. In either case, given both $\|CH_1\|^2, \|DH_2\|^2$ are rational, so is the cross product $\|CH_1\| \cdot \|DH_2\|$. Finally, $\frac{CH_1}{DH_2} = \frac{CH_1^2}{CH_1 \cdot DH_2}$ is rational, as desired.

- B1** Let $\mathcal{P}$ be the set of vectors defined by

$$\mathcal{P} = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \mid 0 \leq a \leq 2, 0 \leq b \leq 100, \text{ and } a, b \in \mathbb{Z} \right\}.$$

Find all $\mathbf{v} \in \mathcal{P}$ such that the set $\mathcal{P} \setminus \{\mathbf{v}\}$ obtained by omitting vector $\mathbf{v}$ from $\mathcal{P}$ can be partitioned into two sets of equal size and equal sum.

Answer. $\begin{pmatrix} 1 \\ 2k \end{pmatrix}$ where $k = 0, 1, \dots, 50$.

Solution. The sum of $\mathcal{P}$ is

$$\begin{pmatrix} 101(0+1+2) \\ 3(0+1+\dots+100) \end{pmatrix} = \begin{pmatrix} 303 \\ 15150 \end{pmatrix}$$

and we require the sum of $\mathcal{P} \setminus \{v\}$ to have both coordinates even (for the partition to be possible). This means if $v = \begin{pmatrix} a \\ b \end{pmatrix}$ we will need $a = 1$ and b even.

Now we claim that such split is possible for any v . First, we note that $|\mathcal{P} \setminus \{v\}| = 101 \times 3 - 1 = 302$, so each split has to have size 151. We'll now fix the following for the first part:

- 26 elements in the form $\begin{pmatrix} 0 \\ x \end{pmatrix}$
- 26 elements in the form $\begin{pmatrix} 2 \\ y \end{pmatrix}$
- 99 elements in the form $\begin{pmatrix} 1 \\ z \end{pmatrix}$

which gives the total first coordinate sum as $26 \times 2 + 99 = 151$, and it has 151 elements too. This way, the second part will have 151 elements with first coordinate sum $75 \times 2 + 1 = 151$.

Let $v = \begin{pmatrix} 1 \\ b \end{pmatrix}$ with b even. The total second coordinate sum is now $3(0 + 1 + \dots + 100) - b = 15150 - b$, so we're targetting the second coordinate sum $7575 - \frac{b}{2}$. For the 99 elements in the first part $\begin{pmatrix} 1 \\ z \end{pmatrix}$ we take z in every possible combination except $z = b$ and $z = 100 - b$ (which does give 99 elements), so we have the total sum as $(0 + 1 + \dots + 100) - (b + 100 - b) = 4950$.

Now for each of the 26 terms in terms of $\begin{pmatrix} 0 \\ b \end{pmatrix}$, the minimum second coordinate sum is

$$0 + 1 + \dots + 25 = 325$$

and the maximum second coordinate sum is

$$75 + 76 + \dots + 100 = 2275$$

Now consider integers $0 \leq x_1 < x_2 < \dots < x_{26} \leq 100$, so long as x_i aren't at their maximum (i.e. not all x_i has $x_i = 100 - (26 - i)$), there always exists i with either $x_{i+1} - x_i \geq 2$ or $i = 26$ and $x_i < 100$. This means we can always add 1 to x_i and increase the sum by 1. This means that in this setup, all numbers in the range $[325, 2275]$ are attainable. Similar situation goes for the 26 entries with first coordinate 2.

Thus, the second coordinate sum can now range between

$$[4950 + 2 \times 325, 4950 + 2 \times 2275] = [5600, 9500]$$

so every integer in this range is attainable. In particular, given that $7525 \leq 7575 - \frac{b}{2} \leq 7575$, the target number $7575 - \frac{b}{2}$ is attainable. Thus using this setup, the other part will also have sum $7575 - \frac{b}{2}$.

B2 Let n be a positive integer, and let $f_n(z) = n + (n-1)z + (n-2)z^2 + \dots + z^{n-1}$. Prove that f_n has no roots in the closed unit disk $\{z \in \mathbb{C} : |z| \leq 1\}$.

Solution. If z were to be a root to f_n , then z is also a root of $f_n(z)(z-1) = z^n + z^{n-1} + \dots + z - n$, i.e. $n = z^n + z^{n-1} + \dots + z$. If $|z| \leq 1$, then

$$n = |z^n + z^{n-1} + \dots + z| \leq |z^n| + \dots + |z| \leq 1 + \dots + 1 = n$$

so all the equalities above must hold, i.e. we need both $|z^n + z^{n-1} + \dots + z| = |z^n| + \dots + |z|$ and $|z^n| + \dots + |z| = 1 + \dots + 1$. The second condition implies that $|z| = 1$, so in particular z cannot be 0. The first condition implies that $\frac{z^k}{z^{k-1}} = z$ must be a positive real number, so these conditions together implies that $z = 1$ is the only possibility. But then $f_n(1) = \frac{n(n+1)}{2} > 0$, contradiction.

- B3** Find all positive integers $n < 10^{100}$ for which simultaneously n divides 2^n , $n - 1$ divides $2^n - 1$, and $n - 2$ divides $2^n - 2$.

Answer. 2^{2^m} for $m = 0, 1, 2, 3$.

Solution. We first show that all numbers are in this form. The first condition $n \mid 2^n$ implies that n cannot have prime divisors other than 2, so $n = 2^k$ for some k . Notice, also that this is also a sufficient condition because $\frac{2^{2^k}}{2^k} = 2^{2^k-k}$ with $2^k > k$ all the times, so the quotient is indeed an integer.

Next, $n - 1 \mid 2^n - 1$ can be written as $2^k - 1 \mid 2^{2^k} - 1$. Using the fact that $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1$, (or equivalently, $2^a - 1 \mid 2^b - 1$ iff $a \mid b$), we have $k \mid 2^k$, and therefore $k = 2^\ell$ for some integer ℓ .

Finally, $n - 2 \mid 2^n - 2$ also implies $2^k - 2 \mid 2^{2^k} - 2$; with $k = 2^\ell$, $k > 0$ so we can divide both sides by 2 and obtain $2^{k-1} - 1 \mid 2^{2^k-1} - 1$. By the same logic as above, $k - 1 \mid 2^k - 1$ and since $k = 2^\ell$, we have $2^\ell - 1 \mid 2^{2^\ell} - 1$, so $\ell \mid 2^\ell$ by the same logic. Thus $\ell = 2^m$. We now have $n = 2^{2^{2^m}}$ for $m \geq 0$.

Finally, we can verify that such n works. To find those with $n < 10^{100}$, notice that

$$2^{300} = (2^3)^{100} = 8^{100} < 10^{100} < 16^{100} = (2^4)^{100} = 2^{400}$$

and therefore $2^{2^m} < 400$. Since $400 < 2^9$, we also need $2^m < 9$, so $m \leq 3$. On the other hand, when $n = 2^{2^{2^3}}$ we have $n = 2^{2^{2^3}} = 2^{2^8} = 2^{256} < 2^{300} < 10^{100}$ so $m = 3$ works.

- B4** Given a real number a , we define a sequence by $x_0 = 1$, $x_1 = x_2 = a$, and $x_{n+1} = 2x_nx_{n-1} - x_{n-2}$ for $n \geq 2$. Prove that if $x_n = 0$ for some n , then the sequence is periodic.

Solution. We first show that we can eliminate the case where $|a| > 1$. Indeed, assume that $|a| > 1$. Let's show that $|x_n|$ is nondecreasing. Indeed, if, for some n , we have $|x_n| \geq |x_{n-1}| \geq |x_{n-2}|$ then

$$\begin{aligned} |x_{n+1}| &= |2x_nx_{n-1} - x_{n-2}| \\ &\geq |2x_nx_{n-1}| - |x_{n-2}| \\ &\geq |2x_nx_{n-1}| - |x_{n-1}| \\ &= |x_{n-1}|(|2x_n| - 1) \\ &\geq |2x_n| - 1 \\ &\geq 2|x_n| - |x_n| \\ &\geq |x_n| \end{aligned}$$

where we used the fact that $|x_n| \geq |x_{n-1}| \geq 1$. Thus with the base case that $|a_0| \leq |a_1| \leq |a_2|$ (as $1 \leq a \leq a$), we have $|x_n|$ nondecreasing, and thus cannot reach 0 at all.

Now that $|a| < 1$, we can write $a = \cos \theta$ and show by induction that $x_n = \cos(F_n \theta)$ with F_n being the n -th Fibonacci number with $F_0 = 0$, $F_1 = F_2 = 1$. Again we can proceed by induction: base case is given by the initial problem condition. Now suppose that $x_k = \cos(F_k \theta)$ for all $1 \leq k \leq n$ for some n . Then (using $F_{n-2} + F_{n-1} = F_n$)

$$\begin{aligned} x_{n+1} &= 2x_nx_{n-1} - x_{n-2} \\ &= 2\cos(F_n\theta)\cos(F_{n-1}\theta) - \cos(F_{n-2}\theta) \\ &= 2\cos(F_n\theta)\cos(F_{n-1}\theta) - \cos((F_n - F_{n-1})\theta) \\ &= 2\cos(F_n\theta)\cos(F_{n-1}\theta) - (\cos(F_n\theta)\cos(F_{n-1}\theta) + \sin(F_n\theta)\sin(F_{n-1}\theta)) \\ &= \cos(F_n\theta)\cos(F_{n-1}\theta) - \sin(F_n\theta)\sin(F_{n-1}\theta) \\ &= \cos(F_{n+1}\theta) \end{aligned}$$

as desired.

Having established this, let n_0 be such that $x_{n_0} = 0$. Then $0 = x_{n_0} = \cos(F_{n_0}\theta)$, meaning that $F_{n_0}\theta = (\frac{2k_0+1}{2})\pi$ for some integer k_0 . This also means that $\theta = (\frac{2k_0+1}{2F_{n_0}})\pi$. We first recall that for any integer m , the Fibonacci sequence is periodic modulo m : indeed, if we write the Fibonacci sequence modulo m and consider all pairs (F_n, F_{n+1}) , then there are at most m^2 distinct elements from the pairs, and thus there are some that are the same. If $(F_k, F_{k+1}) = (F_\ell, F_{\ell+1})$ modulo m for some $k < \ell$, then we can inductively show (both forward and backward) that $F_{k+x} = F_{\ell+x}$ for any integer (positive or negative) x . Now if p_0 is the period of the Fibonacci sequence when $m = 4F_0$ then $F_{k+p_0} \equiv F_k \pmod{4F_0}$ for all k , and therefore $(F_{k+p_0} - F_k)\theta = (F_{k+p_0} - F_k)(\frac{2k_0+1}{2F_{n_0}})\pi$ is an integer multiple of 2π . Since $\cos$ is periodic modulo 2π , we have $\cos(F_{k+p_0}\theta) = \cos(F_k\theta)$, and thus $x_k = x_{k+p_0}$.

B5 Let $f = (f_1, f_2)$ be a function from $\mathbb{R}^2$ to $\mathbb{R}^2$ with continuous partial derivatives $\frac{\partial f_i}{\partial x_j}$ that are positive everywhere. Suppose that

$$\frac{\partial f_1}{\partial x_1} \frac{\partial f_2}{\partial x_2} - \frac{1}{4} \left(\frac{\partial f_1}{\partial x_2} + \frac{\partial f_2}{\partial x_1} \right)^2 > 0$$

everywhere. Prove that f is one-to-one.

Solution. Suppose that $f(x_0, y_0) = f(x_0 + a, y_0 + b)$ for some $a, b \in \mathbb{R}^2$. We consider the following: if $g_1(z) = f_1(x_0 + za, y_0 + zb)$ then $g'_1(z) = a \frac{\partial f_1}{\partial x_1}(x_0 + za, y_0 + zb) + b \frac{\partial f_1}{\partial x_2}(x_0 + za, y_0 + zb)$, then

$$\begin{aligned} 0 &= f_1(x_0 + a, y_0 + b) - f_1(x_0, y_0) \\ &= g_1(1) - g_1(0) \\ &= \int_0^1 g'_1(z) dz \\ &= \int_0^1 a \frac{\partial f_1}{\partial x_1}(x_0 + za, y_0 + zb) + b \frac{\partial f_1}{\partial x_2}(x_0 + za, y_0 + zb) dz \\ &= a \int_0^1 \frac{\partial f_1}{\partial x_1}(x_0 + za, y_0 + zb) dz + b \int_0^1 \frac{\partial f_1}{\partial x_2}(x_0 + za, y_0 + zb) dz \end{aligned}$$

and similarly, $a \int_0^1 \frac{\partial f_2}{\partial x_1}(x_0 + za, y_0 + zb) dz + b \int_0^1 \frac{\partial f_2}{\partial x_2}(x_0 + za, y_0 + zb) dz = 0$. Denote $x(z) = x_0 + za$ and $y(z) = y_0 + zb$, and the matrix $\begin{pmatrix} \int_0^1 \frac{\partial f_1}{\partial x_1}(x(z), y(z)) dz & \int_0^1 \frac{\partial f_1}{\partial x_2}(x(z), y(z)) dz \\ \int_0^1 \frac{\partial f_2}{\partial x_1}(x(z), y(z)) dz & \int_0^1 \frac{\partial f_2}{\partial x_2}(x(z), y(z)) dz \end{pmatrix}$ as M , we have the following:

$$M \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We claim that M is invertible, thereby showing that $\begin{pmatrix} a \\ b \end{pmatrix}$ must be $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and thus f is injective. This is the same as showing that the determinant of M is nonzero. We will in fact show that it is positive. On one hand, the integral Cauchy-Schwarz inequality says that

$$\int_0^1 \frac{\partial f_1}{\partial x_1}(x(z), y(z)) dz \int_0^1 \frac{\partial f_2}{\partial x_2}(x(z), y(z)) dz \geq \left(\int_0^1 \sqrt{\frac{\partial f_1}{\partial x_1}(x(z), y(z)) \frac{\partial f_2}{\partial x_2}(x(z), y(z))} dz \right)^2$$

(we are allowed to use square roots since all partial derivatives are positive). On the other

hand using the inequality $AB \leq \frac{(A+B)^2}{4}$ for all $A, B \in \mathbb{R}$ we also have

$$\begin{aligned} \int_0^1 \frac{\partial f_1}{\partial x_2}(x(z), y(z)) dz \int_0^1 \frac{\partial f_2}{\partial x_1}(x(z), y(z)) dz &\leq \frac{(\int_0^1 \frac{\partial f_1}{\partial x_2}(x(z), y(z)) dz + \int_0^1 \frac{\partial f_2}{\partial x_1}(x(z), y(z)) dz)^2}{4} \\ &= \frac{(\int_0^1 \frac{\partial f_1}{\partial x_2}(x(z), y(z)) + \frac{\partial f_2}{\partial x_1}(x(z), y(z)) dz)^2}{4} \\ &= \left(\int_0^1 \frac{1}{2} (\frac{\partial f_1}{\partial x_2}(x(z), y(z)) + \frac{\partial f_2}{\partial x_1}(x(z), y(z))) dz \right)^2 \end{aligned}$$

The given inequality implies that for all $z \in [0, 1]$ we have (moving one term to the other side and take square roots on both sides)

$$\sqrt{\frac{\partial f_1}{\partial x_1}(x(z), y(z)) \frac{\partial f_2}{\partial x_2}(x(z), y(z))} > \frac{1}{2} \left(\frac{\partial f_1}{\partial x_2}(x(z), y(z)) + \frac{\partial f_2}{\partial x_1}(x(z), y(z)) \right)$$

and by the continuity of partial derivatices we have

$$\int_0^1 \sqrt{\frac{\partial f_1}{\partial x_1}(x(z), y(z)) \frac{\partial f_2}{\partial x_2}(x(z), y(z))} dz > \int_0^1 \frac{1}{2} \left(\frac{\partial f_1}{\partial x_2}(x(z), y(z)) + \frac{\partial f_2}{\partial x_1}(x(z), y(z)) \right) dz$$

and hence

$$\int_0^1 \frac{\partial f_1}{\partial x_1}(x(z), y(z)) dz \int_0^1 \frac{\partial f_2}{\partial x_2}(x(z), y(z)) dz > \int_0^1 \frac{\partial f_1}{\partial x_2}(x(z), y(z)) dz \int_0^1 \frac{\partial f_2}{\partial x_1}(x(z), y(z)) dz$$

and therefore

$$\det(M) = \int_0^1 \frac{\partial f_1}{\partial x_1}(x(z), y(z)) dz \int_0^1 \frac{\partial f_2}{\partial x_2}(x(z), y(z)) dz - \int_0^1 \frac{\partial f_1}{\partial x_2}(x(z), y(z)) dz \int_0^1 \frac{\partial f_2}{\partial x_1}(x(z), y(z)) dz > 0$$